

CueFit

AI-Powered Home Physiotherapy Platform

Bringing clinical supervision home, one rep at a time

Event

Fanar Hackathon 2026

Team

See Sharp

Track

Healthcare Information Systems &
Patient Support

Problem Statement

Home physiotherapy lacks the supervision patients need to recover safely and effectively.



What goes wrong at home



No Supervision

No one to check if exercises are done correctly.



Incorrect Form

Compensation patterns develop silently and can worsen injuries.



Low Motivation

Lack of engagement and accountability leads to poor adherence.



No Progress Tracking

No way to measure improvement or know if the program is working.

Up to 70% of patients may not adhere to prescribed home exercise programs, depending on the condition and population studied ([link](#)).

Solution - CueFit

Integrated computer-vision pose tracking and bilingual AI coaching for home-based clinical supervision.



Exercise Monitor

- ✓ Webcam-based 3D landmark tracking using MediaPipe.
- ✓ Real-time measurement of joint angles and ROM.
- ✓ Automated rep counting and symmetry scoring.
- ✓ Detection of compensation patterns (e.g., knee valgus).



AI Coach Agent

- ✓ Bilingual conversational agent (English & Arabic).
- ✓ Interprets movement data into actionable cues.
- ✓ Hands-free "Hey Coach" wake-word interaction.
- ✓ Automated clinical session reporting and feedback.

TYPICAL PATIENT SESSION LIFECYCLE

STEP 01



Select Workout

Choose from regional workout sets.

STEP 02



Review Form

Guided instructions and target goals.

STEP 03



Active Session

Live pose tracking and skeletal overlay.

STEP 04



Hands-free Q&A

Ask "Hey Coach" for real-time help.

STEP 05



Feedback

Personalized spoken performance data.

STEP 06

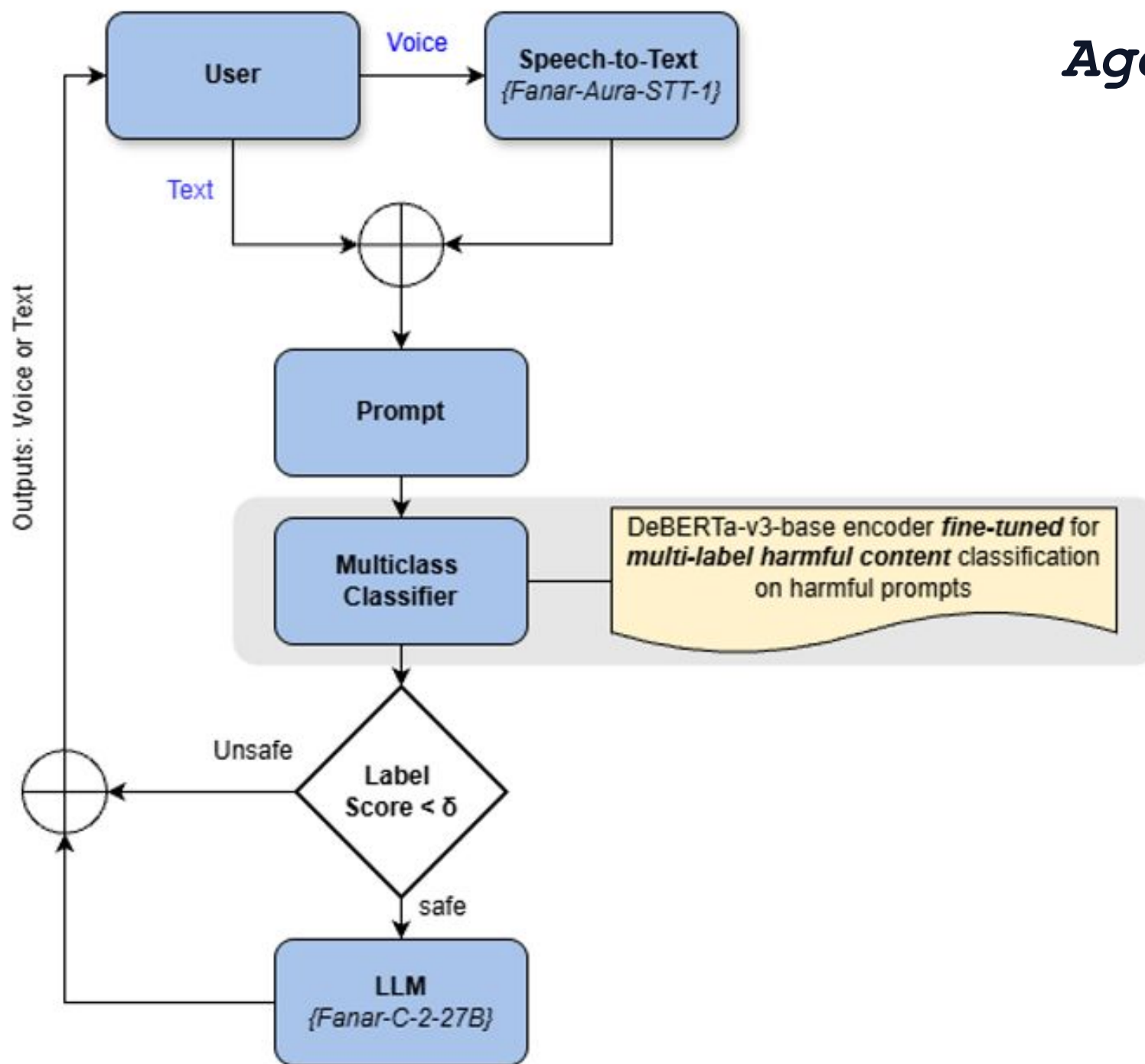


AI Report

Detailed breakdown of the session.



Agentic Workflow - The AI Coach



Computer Vision Pipeline

PIPELINE PERFORMANCE

Processing Rate

~5 FPS

Joint Landmarks

33 Points

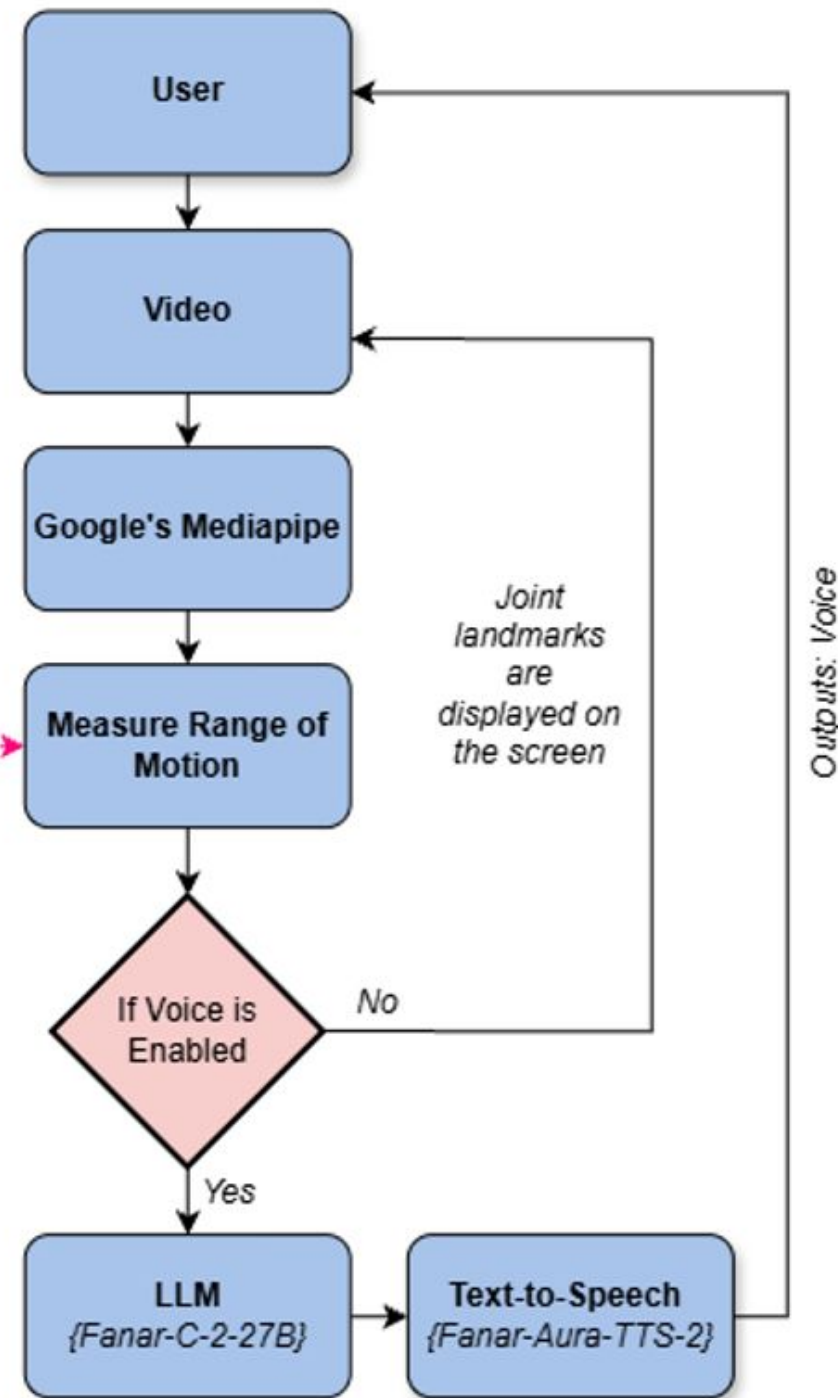
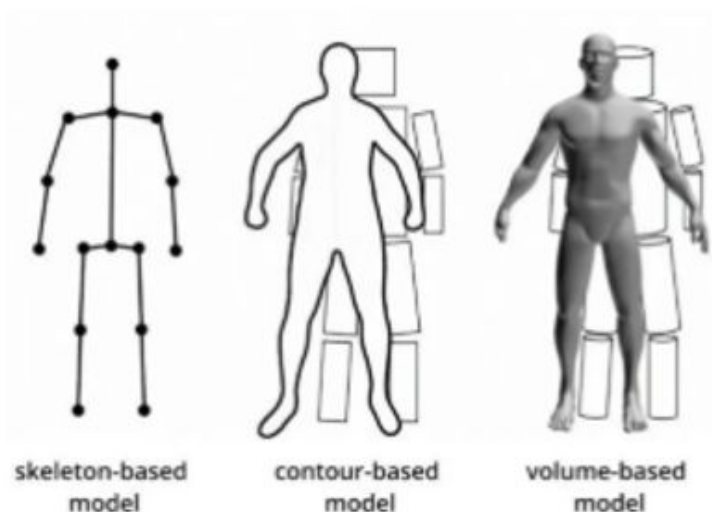
Confidence Gate

> 0.50

Tracked Angles

10 Total

* CueFit utilizes a **skeleton-based model** (left in the diagram) to ensure high-speed clinical biomechanics without high compute overhead.



Conversation Safety Layer

An independent conversation-aware safety layer that intercepts every request before it reaches Fanar. This not only reduces unnecessary in

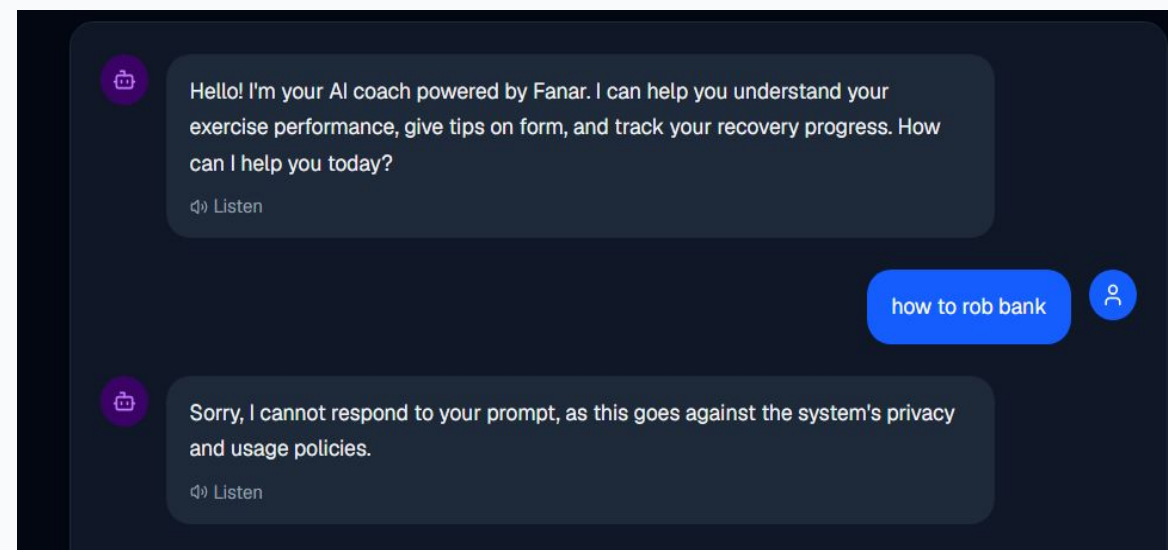
Moderation Classifier



Confidence Threshold
Gating

Polite Refusal
Injection

Pre-LLM
Filtering



Evaluation

- 3 sessions attempting contextual jailbreak
- Fanar blocked 2/3 sessions.
- Safety layer blocked all 3 sessions.

| Future Plans & Fanar Recommendations

Therapist dashboard

Develop a remote portal where clinicians review session reports and adjust programmes.

Mobile support

Use phone camera instead of webcam. We recommend Fanar expose a **WebSocket-based streaming ASR endpoint**; with it, the voice coach becomes genuinely hands-free and low-latency on mobile.

Real-time interruption

Currently the patient has to wait for the coach to finish speaking. With **streaming ASR and streaming TTS** from Fanar, the patient could say "stop" or "repeat that" mid-sentence and the coach would respond immediately — a far more natural physiotherapist-like dynamic that we cannot build on the current batch API alone.

| Closing

CueFit

AI Physiotherapy

**Bringing clinical supervision home,
one rep at a time.**

Bridging the supervision gap with computer vision and agentic AI to ensure safe, personalized, and effective recovery for every patient.

Built for the Fanar Hackathon · June 2026

